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SOLVIX

Management of Information Technologies

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Table of Contents

Introduction.....	4
Project Description.....	5
Personas.....	8
User stories and prototyping.....	15
User Interviews and Prototyping.....	19
Usability Testing Plan.....	22
Scrum.....	33
Project stakeholders.....	36
Engineering Management – Leadership.....	38
Final Application Showcase.....	44
Client Portal.....	49
Complaint Workflow.....	52
System Features.....	53
Conclusion.....	53

Introduction

This document presents the complete work carried out by our team throughout the Management of Information Technologies course. The project, named Solvix, is a web-based complaint management platform developed for Zalando, one of Europe's largest e-commerce retailers.

The following chapters cover every stage of the project lifecycle: from the initial project description and stakeholder analysis, through user research and prototyping, to Agile sprint execution and engineering management principles. Each chapter builds upon the previous one, forming a coherent record of our team's progress and decisions.

The document is structured to reflect both the technical and organisational dimensions of the project, and concludes with a summary of outcomes, usability findings, and recommendations for future development.

Project Description

Solvix is a simple online platform that helps Zalando manage customer complaints in a clear and organized way.

It keeps all complaints in one place, makes communication between departments easier, and helps employees resolve issues faster. The system is also beneficial for customers, as it allows them to submit complaints easily, track their status, and receive quicker responses. This improves both company efficiency and customer satisfaction.

GitHub: <https://github.com/simona-cholakova/Solvix>

1. Project Client

ZALANDO - Online Retail Company



2. Importance

As a large European e-commerce company, Zalando processes thousands of customer orders daily. Complaints related to delivery delays, returns, refunds, and damaged products require structured and efficient handling.

Implementing a Complaint Management System will:

- Improve customer satisfaction
- Reduce complaint resolution time
- Improve internal communication between departments
- Strengthen customer trust and brand reputation

3. Purpose & objectives

- **Purpose:** To develop a web-based system for managing and tracking customer complaints.
- **Objectives:**
 - Enable customers to submit complaints online
 - Categorize complaints by type (refunds, late delivery, damaged product...)
 - Assign complaints to responsible departments
 - Allow status tracking (Pending, In Progress, Resolved)
 - Generate management reports

- Reduce resolution time

4. Possible Requirements of the Contracting Authority

- **Description of the Methodology:** The project will follow Agile methodology with short development cycles and continuous feedback.
- **Techniques & technologies used:**
 - Frontend: React
 - Backend: Node.js / Express
 - Database: MySQL
 - REST API integration
- **Description of the Finished Product:** The final system will include:
 - Complaint submission form
 - Admin dashboard
 - Complaint status tracking
 - Department assignment feature
 - Basic analytics and reporting module
- **Reporting and Final Report:**
 - Weekly progress reports
 - Final technical and functional documentation
 - Testing and evaluation report

5. Project Limitations (Options)

- **Rough Schedule (3 months total):**
 - Month 1:
 - Requirements analysis
 - System design
 - Database design
 - Month 2:
 - Frontend and backend development
 - API integration
 - Month 3:
 - Testing
 - Bug fixing
 - Deployment and documentation

→ **Available Funds:**

- Estimated budget: **€30,000**
- *Sponsor: **ZPS - Zveza potrošnikov Slovenije***

→ **Milestones:**

- Requirements approved – End of Week 2
- Prototype ready – End of Month 2
- Final release - End of Month 3

6. Project Management

→ **Project Manager:** Simona Cholakova

→ **Team Members:**

- Backend Developer - Jaka Hrvatin
- Frontend Developer - Nade Belovinova
- QA Tester - Ema Čikotić

7. Date of Order: 21.02.2026

8. Client: Zalando

9. Signature:

Personas

Persona 1:



Demographic

Age	45
Gender	Female
Marital Status	Married
Income	€70.500
Experience	18 years
Employment type	Full-time
Digital Skill Level	Advanced

Anna Wattson

Customer Care Manager
at Zalando

Quotes:

“With the right organization, every complaint leads to growth.”

Screening question: Do you manage customer complaints, assign them to departments, track their status, and use digital tools to improve resolution and team performance?

Function: Customer Care Manager at Zalando (Responsible for complaint handling and team performance)

About Anna:

Anna Wattson is the Customer Care Manager at the headquarters of Zalando in Berlin, Germany. She goes to work Monday to Friday from 9am to 5pm. She oversees the daily work of the customer service team. She makes sure customer complaints and problems are handled quickly and properly. She approves the goals set by the team to ensure customer satisfaction and works closely with them to consistently achieve those goals.

Anna's main focus is monitoring and lowering complaint resolution time, keeping track of customer satisfaction scores, and team performance metrics. Anna reviews complex complaints, makes final decisions on refunds or compensations, and ensures consistent communication with customers and maintains the company's reputation.

Anna is flexible, enthusiastic and eager to make her career number one priority in her life. She is always keen on new technology that makes her work life easier and more beneficial for her team and the customers. She believes that excellent customer service is key to business success and that learning from complaints helps improve future performance.

Thinks	Anna wants complaints to be handled quickly, efficiently, and fairly . She believes the current process could be more organized and transparent. She thinks any new technology should be easy to use and not hard to learn , so her team can adopt it quickly.
Sees	She monitors team dashboards, reports, and past complaint data to see patterns and bottlenecks. She learns about improvements from industry best practices, other e-commerce companies, and internal reviews .
Feels	Anna feels frustrated when complaints pile up or resolution is delayed and especially when she doesn't know the current status of the complaints. She feels satisfied and motivated when her team resolves complaints efficiently and customers leave positive feedback. She wants to empower her team and make their work easier and more efficient.
Does	She reviews escalated complaints daily, where some of them have no structure at all. She assigns complaints to the right departments and monitors their status. She generates reports to analyze recurring issues and team performance. She communicates with team members to provide guidance on complaint resolution.

Problem Scenarios	Current Alternatives	Value Proposition
"I need a clear overview of all customer complaints and their current status."	I use spreadsheets, emails, and manual tracking to monitor complaints. This takes time, information gets lost, and it's hard to see real-time progress.	Keep all complaints in one centralized dashboard with real time status tracking, making monitoring faster, clearer, and more organized.
"I need to make sure complaints are assigned to the right department quickly."	I manually forward emails or contact departments separately. This causes delays, miscommunication, and sometimes duplicated work.	Allow direct department assignment within the system, improving internal communication and reducing resolution time.
"I need to reduce complaint resolution time and improve customer satisfaction."	I manually review complaint records and follow up with departments to check progress. This delays decision-making and makes it hard to identify bottlenecks quickly.	Offer real time complaint tracking and insightful analytics, enabling to quickly detect delays, lower response times, and therefore enhance customer satisfaction.

Persona 2:



Lukas Weber
Accountant at Zalando

Demographic

Age	34
Gender	Male
Marital Status	Single
Income	€50.000
Experience	7 years
Employment type	Full-time
Digital Skill Level	Mediocre

Quotes:

“Returns are part of the business model. But when they hit after the cutoff, they hit my sleep too.”

Screening question: Does your role involve handling or adjusting financial reports due to customer complaints, refunds, or returns?

Function: Financial Accountant responsible for revenue reporting and return adjustments

About Lukas:

Lukas Weber is a Financial Accountant at an e-commerce company similar to Zalando. He is responsible for revenue reporting and month-end closing, where customer returns and complaints directly impact financial results. Lukas often feels stressed when late returns or refund requests arrive after reporting deadlines, forcing him to make last-minute adjustments. He values accuracy, predictability, and better visibility into complaint trends to reduce financial volatility and pressure during closing periods.

Lukas prefers structured systems and clear communication between departments to avoid last-minute financial adjustments. He values transparency and early visibility into refunds and complaints so he can prepare accurate reports. Better access to real-time complaint data would reduce stress during month-end closing and improve financial forecasting.

Thinks	Lukas wants to know about complaints and returns as soon as possible so he can re-adjust finances.
Sees	He sees late customer complaints that enter the system after deadlines, making reports less accurate.
Feels	Stressed during the month closing reports, because he anticipates last

	minute changes and is frustrated that he does not have insight into them sooner.
Does	Contacts customer service to clarify any uncertainties and build an excel report to estimate return provisions.

Problem Scenarios	Current Alternatives	Value Proposition
“It’s the last day of the month, and Lukas has already closed the books. Two days later, a batch of delayed customer complaints is processed by customer service. Many of them include refund approvals for high-value items.”	At the end of every month I contact the Customer service department to ask them about any new complaints.	Keep all complaints in one centralized dashboard with real time status tracking, making monitoring faster, clearer, and more organized.
“During month-end closing, Lukas receives multiple emails from customer service about unresolved complaints. The information is incomplete, spread across systems, and not standardized.”	I manually format the information provided to me and then handle the accounting after.	Keep all complaints in one place and already formatted to reduce manual overhead.

Persona 3:

Demographic



Age	19
Gender	Female
Marital Status	Single
Income	€500(monthly allowance)
Employment type	Student (Not employed)
Digital Skill Level	High

Dorothy Novak

Quotes:

First year student in Slovenia

“If something goes wrong, fixing it shouldn’t take hours.”

Screening question: Have you ever experienced a technical error with a promotional code and sought a quick way to report it and track the resolution?

Function: Zalando customer (student shopper)

About Dorothy:

Dorothy is a first year student at the Faculty of Arts in Ljubljana, Slovenia.

Like most students, she lives on a tight budget and treats herself to new clothes, shoes or accessories only when there is a good sale or a discount code available. She is a frequent Zalando shopper because of the variety and the convenience of home delivery. Recently, she received a promotional code via email that explicitly stated it was valid for the next 15 days.

She spent her evening carefully picking out a pair of sneakers and a jacket, planning to purchase them the next morning after the lecture. However, when she tried to apply the code less than 24 hours later, the system rejected it as “expired”.

For Dorothy this isn’t just a technical glitch; it’s the difference between being able to afford the items or having to empty her cart. She is frustrated and feels misled by the marketing, but she doesn’t want to spend hours on a phone line or send redundant emails. She wishes the fix to her problem was just a few clicks away and didn’t require many social interactions.

Thinks	Dorothy thinks that large companies should honor their advertisements. She believes that if a system error occurs, the company should provide a replacement code immediately. She wants a platform where she will know that her complaints will be immediately heard and acted upon.
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Sees	She sees influencers posting about “limited time” deals and feels pressure to shop fast, but she also sees “broken” customer service links that lead nowhere.
Feels	Feels annoyed and “cheated” out of a deal. As a student, she is price sensitive, so the loss of a 30% discount feels significant. She feels anxious about whether she will get a response before the items she wants sell out.
Does	She takes a screenshot of the original email (showing the 15 day validity) and the error message in her cart. She looks for a “Contact Us” or “Submit Complaint” button that doesn’t require a long phone call.

Problem Scenarios	Current Alternatives	Value Proposition
“I have proof that my code should work, but the checkout page says it’s expired. I need to show someone my screenshot”	Searching for a hidden customer service email or waiting in a live chat queue for 40 minutes.	Quickly upload her screenshot via a dedicated complaint form, ensuring her evidence is seen by the right department.
“I submitted a complaint through email, but I have no idea if anyone is looking at it. I’m afraid the sneakers will go out of stock.”	Refreshing email inbox constantly and hoping for a non-automated reply.	Use the Status Tracking feature to see that her complaint is “In Progress”, giving her peace of mind that resolution is coming.

User stories and prototyping

Persona 1 : Anna Watson, the Customer Care Manager

Epic Story: As a team leader, I want one simple screen to see and assign all incoming complaints, so I can make sure my team is working on the right things and no customer is left waiting.

User story	Test cases
As a Customer Care Manager, I want to see all customer complaints in a centralized dashboard so that I can quickly monitor their status and manage them efficiently.	TC1: Make sure all complaints are displayed in a dashboard. TC2: Make sure each complaint shows ID, customer name, complaint type, status, department and submission date. TC3: Make sure complaints can be filtered by status, department, or complaint type.
As a Customer Care Manager, I want to assign complaints to the appropriate department so that issues are handled by the correct team.	TC1: Make sure a department dropdown is available when viewing a complaint. TC2: Make sure the assignment can be saved. TC3: Make sure the complaint appears in the assigned department's queue.
As a Customer Care Manager, I want to track the status of complaints so that I can ensure they are resolved within an acceptable time.	TC1: Make sure complaint status is visible (Pending, In Progress, Resolved). TC2: Make sure departments can update complaint status. TC3: Make sure the resolution time is recorded and displayed.

Persona 2: Lukas Webber, the accountant

Epic story: As a financial accountant, I want to have full transparency and structured access to all financially relevant complaints, so that I can eliminate manual tracking, reduce stress during month end closing and ensure the company's financial reports are 100% accurate.

User story	Test cases
As a Financial Accountant,I want to view a real time list of complaints categorized as "Refunds" or "Returns" (or else), so that I can identify potential financial liabilities before they hit the books.	TC1: Verify that the dashboard allows filtering by "Complaint Type" (e.g., Refund, Damaged Product). TC2: Ensure that each financial complaint displays the specific Order Value and Requested Refund Amount. TC3: Confirm that the dashboard status updates automatically when a Customer Care agent approves a refund. TC4: Verify that a "Closing Date" filter exists, allowing Lukas to see only complaints submitted within the current fiscal month.
As a Financial Accountant, I want to export complaint data into a structured CSV or Excel file, so that I can import the data directly into my accounting software without manual re-formatting.	TC1: Verify the existence of an "Export to CSV/Excel" button within the reporting module. TC2: Confirm that the exported file contains the following mandatory columns: Complaint_ID, Status, Amount, Date_Approved, and Order_ID. TC3: Ensure that the export includes a "Status" flag (e.g., "Pending" vs "Resolved") to help differentiate actual payouts. TC4: Verify that the system generates a success message or notification once the file is ready for download.
	TC1: Verify that the reporting module includes a "Monthly Refund Summary"

As a Financial Accountant, I want to receive a summary report of all approved refunds for the current month, so that I can easily reconcile them with the financial statements.	option. TC2: Ensure the report lists Complaint ID, Order ID, Refund Amount, Approval Date, and Status. TC3: Confirm that Lukas can filter the report by month and year. TC4: Verify that only complaints with status = Resolved/Refund Approved appear in the report.
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Persona 3 : Dorothy Novak, the Student

Epic Story: As a busy student, I want a quick way to report my issue and see exactly what is happening with it, so I can get my discount fixed without wasting time on phone calls or checking my email constantly.

<i>User story</i>	<i>Test cases</i>
I want to upload proof (like screenshots) when submitting a complaint, so that the support team can quickly verify my issue and resolve it.	TC1: The user can attach files to the complaint form. TC2: The file is visible to the support team handling the complaint.
I want to track the status of my complaint, so that I know whether it is being processed and don't have to constantly check my email.	TC1: The complaint has statuses like Submitted, Seen, In-Progress, Resolved . TC2: Users can view the complaint status in their account and receive push notifications whenever the status is updated. TC3: The complaint status updates automatically when support takes action.
As a customer, I want to receive a clear resolution or response to my complaint directly on the platform, so that I don't have to communicate through multiple emails or support channels.	TC1: Verify that the system allows support staff to send a response or resolution message directly within the complaint page . TC2: Ensure that the customer can view the response when they open their complaint in their account . TC3: Confirm that the response clearly shows the resolution details (e.g., replacement promo code, refund approval, explanation). TC4: Verify that the system marks the complaint as "Resolved" once the resolution message is sent . TC5: Ensure the customer can acknowledge or confirm the resolution (for example with a "Mark as solved" button).

User Interviews and Prototyping

People interviewed:

Person	Role	Relation to system
Eva Petrushevska	Zalando user	Represents regular e-commerce users
Darko Cholakov	Accountant	Represents financial staff who need visibility into complaints
Alja Eremic	Zalando user	Represents regular e-commerce users

Interview 1 - Accountant:

Name: Darko Cholakov

Age: 40

Position: Financial Accountant at a mid-sized online retail company

Experience: 6 years in accounting

Questions & Answers

Q1: Do customer complaints affect your financial reporting?

A: Yes, complaints that lead to refunds or returns directly impact revenue reporting and monthly financial statements.

Q2: How do you currently receive information about refund complaints?

A: Usually through emails or Excel files sent by the customer service team.

Q3: What problems do you experience with the current process of receiving complaints?

A: Information sometimes arrives late or in different formats, which requires manual adjustments.

Q4: Would access to a centralized system with all complaints help your work?

A: Yes, it would make it easier to track refunds and prepare financial reports, only if the learning curve is easy enough to accommodate quickly.

Q5: What information about complaints is most important for financial analysis?

A: Complaint type, price, refund amount, and approval date.

Q6: Would the ability to filter complaints by type (refund, return, damaged product) be useful?

A: Yes, especially when analyzing refunds, because they affect my financial reports directly..

Q7: Would exporting complaint data into Excel or CSV be useful for you?

A: I think that any solution that would have a structured answer would be better than the current alternatives that we have.

Q8: Would search functionality help you find specific complaints or orders faster?

A: Yes, especially when checking specific orders during financial audits or reporting.

Interview 2 – Zalando User

Name: Eva Petrushevskia

Age: 21

Occupation: University student

Shopping habits: Frequently shops online and uses discount codes

Questions & Answers

Q1: Have you ever experienced a problem with an online order?

A: Happily it did not happen to me, but I see that it is a common thing that happens to people online.

Q2: If you did have a problem how would you report it?

A: I would contact customer support through email or phone number if it is available.

Q3: What would be the most frustrating part of that process?

A: Waiting for a response.

Q4: Would it help if you could track the status of your complaint online?

A: Yes, it would make me feel more at ease since I would be able to see that they actually received it.

Q5: Would uploading screenshots/files when submitting a complaint be useful?

A: Only in cases when showing proof of an issue.

Q6: Would it be useful to search for your complaints in your account instead of scrolling through emails?

A: Yes, that would make it easier to find and track status of complaints.

Interview 3 – Zalando User

Name: Alja Eremic

Age: 22

Occupation: University student

Shopping habits: Frequently shops online and uses discount codes

Questions & Answers

Q1: Have you ever experienced a problem with an online order?

A: Yes, once I received the wrong size of shoes.

Q2: How did you approach the problem?

A: I first checked the help section of the website and then I sent an email, but it wasn't very helpful.

Q3: What is the most frustrating part of that process?

A: Waiting for a response, not knowing if the email address is even correct.

Q4: Would it help if you could track the status of your complaint online?

A: Well, yes but still I would prefer the process itself to be just faster.

Q5: Would uploading screenshots/files when submitting a complaint be useful?

A: Not really, but I guess for some people it would be a nice thing.

Q6: Would it be useful to search for your complaints in your account instead of scrolling through emails?

A: I guess yes, if you have multiple complaints at once.

Conclusions

The interviews confirm that the main hypotheses are valid.

- The **accountant** needs structured complaint data, filtering and search features to handle refunds efficiently and reduce manual work.
- **Zalando users** want faster responses and the ability to search their complaints instead of relying on emails.

Overall, both employees and customers would benefit from a **centralized system**, and the interviews highlighted **search functionality** as an important improvement for the prototype.

Usability Testing Plan

Objective and Methods

The goal of this usability test is to evaluate whether the complaint management interface allows Customer Care Managers to efficiently view, filter, and manage complaints.

Objectives:

- Evaluate whether users can quickly locate specific complaints using search and filters
- Assess how easily users can understand complaint information from the dashboard
- Determine if users can navigate to complaint details without confusion
- Evaluate whether users can update complaint status and assign departments correctly
- Identify usability issues and points of confusion during task completion
- Collect user feedback on clarity, layout, and missing features

Method:

The usability testing will be conducted in person using an interactive Figma prototype. Participants will be asked to complete predefined tasks while thinking aloud.

The tester will observe user behavior, measure task completion time, and record any errors, difficulties, or feedback.

Product Version

We will be using our interactive [Figma Prototype](#)

Subjects

The test will include 3 or more participants who match the target persona (users familiar with dashboards or data management systems).

Screening question

- Have you ever used a dashboard or system to manage tasks, complaints, or data?
- How often do you use such systems? (Daily / Occasionally / Never).

Research Composition

#	Item	Duration (min.)	Notes
1	Intro & Explanation	5	Explain the test goal, confirm participation.
2	Test Tasks	10	Introduce the test scenario and ask participants to complete the assigned tasks.
3	Post-Test Debrief	10	Ask for feedback on the prototype and consent to follow up.

Pre-Session Checklist

#	Item	Notes
1	Have the handwritten questions and tasks ready.	Keep open for quick reference.
2	Confirm the prototype is working.	Test all commands and flows.
3	Confirm note-taking setup and timer.	Take hand written notes during the session and time each task.

Session Design

“Thank you for taking the time to participate in this test. My name is ____, and we are testing an early version of a complaint management interface designed to help Customer Care Managers easily navigate complaints and assign them to the appropriate departments.

During the session, I will ask you to complete a few tasks.

Please think out loud as you go and share any thoughts, suggestions, or difficulties you encounter.

Remember, there are no wrong answers. This is not a test of you, but a way for us to improve the interface and make it more user-friendly.

The session will take around 20 minutes.

Do you have any questions before we begin?"

#	Objective	Duration (min.)	Notes
1	Find a specific complaint using search	1	Task: Find the complaint submitted by <i>Emily Davis</i> using search. Output: Can the user quickly locate a complaint using search?
2	Filter complaints by status	1	Task: Filter complaints to show only <i>Pending</i> ones. Output: Can the user apply filters without confusion?
3	Open a complaint	1	Task: Open complaint C-005 from the list. Output: Can the user easily navigate to the complaint details page?
4	Change complaint status	2	Task: Change the complaint status <i>In Progress</i> Output: Can the user understand and update the complaint status without confusion?
5	Assign department	1	Task: Assign a complaint to a different department. Output: Can the user correctly use the department dropdown?
6	Add internal note	2	Task: Add an internal note to the complaint. Output: is the internal notes feature visible and easy to use?

7	Save changes	1	Task: Save the changes made to the complaint. Output: Does the user notice and use the “Save Changes” button?
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Notes:

Notes will be taken during the session, including observations and participant feedback.

Post-Test Debrief

Thank you very much for your participation. Your feedback will help us improve the product. Before we finish, we have a few final questions:

- Was anything confusing or difficult to use?
- Did any part feel too time-consuming?
- What did you find most useful?
- Was anything tedious or frustrating?
- Was there anything missing that you expected to be included?

That's everything for now. Would it be alright if we contacted you later for follow-up questions or additional testing, if needed?

Test Result

Person 1:

Task	Time	Result	Notes
Search “Emily Davis”	12 sec	Success	Typed full name; search worked smoothly
Filter “Pending”	18 sec	Success	Slight hesitation finding filter icon
Open C-005	9 sec	Success	Clicked row immediately
Change status	32 sec	Partial confusion	Wasn’t understood like a dropdown
Assign department	21 sec	Success	Dropdown clear
Add internal note	41 sec	Success	Comment box easy to find
Save changes	14 sec	Failed initially	Didn’t notice button; expected auto-save

General Feedback:

- “The layout is clean but the Save button blends in too much.”
- “Status change interaction feels different from other systems I use.”

Person 2:

Task	Time	Result	Notes
Search "Emily Davis"	17 sec	Struggled	Tried filtering first instead of search
Filter "Pending"	7 sec	Success	Found filter quickly
Open C-005	14 sec	Success	Scanned list slowly
Change status	48 sec	Confused	Didn't understand status categories
Assign department	25 sec	Struggled	Dropdown label unclear
Add internal note	50 sec	Struggled	Thought notes were in a different tab
Save changes	10 sec	Success	Found button after scanning bottom area

General Feedback:

- "I wasn't sure where to add notes."
- "Status options need clearer wording."

Person 3:

Task	Time	Result	Notes
Search "Emily Davis"	7 sec	Success	Very fast; very easy
Filter "Pending"	10 sec	Success	No issues
Open C-005	6 sec	Success	Smooth
Change status	19 sec	Success	No issues
Assign department	11 sec	Success	Found quickly
Add internal note	50 sec	Success	No issues
Save changes	10 sec	Success	No issues

General Feedback:

- "Everything is straightforward."
- "Save button is fine, but could be more prominent."

Person 4:

Task	Time	Result	Notes
Search "Emily Davis"	7 sec	Success	Immediately navigated to the search bar, no issues encountered.
Filter "Pending"	20 sec	Partial confusion	Was unsure whether to reset the previous search or continue with it.
Open C-005	43 sec	Success (with confusion)	Attempted to open C-005, but the prototype opened C-001
Change status	7 sec	Success	No issues
Assign department	5 sec	Success	No issues
Add internal note	16 sec	Success	Found it easily and was happy to use it.
Save changes	3 sec	Success	No issues

General Feedback:

- "The save button could be placed at the very end of the complaint details, as I might just scroll up the Save button and miss the Internal Notes section."
- "The complaint could be opened by clicking anywhere on the row.."
- "A Reset filter button would be beneficial."
- "Could be useful to have a color coding for each status."
- "It is easy to use."

Conclusion

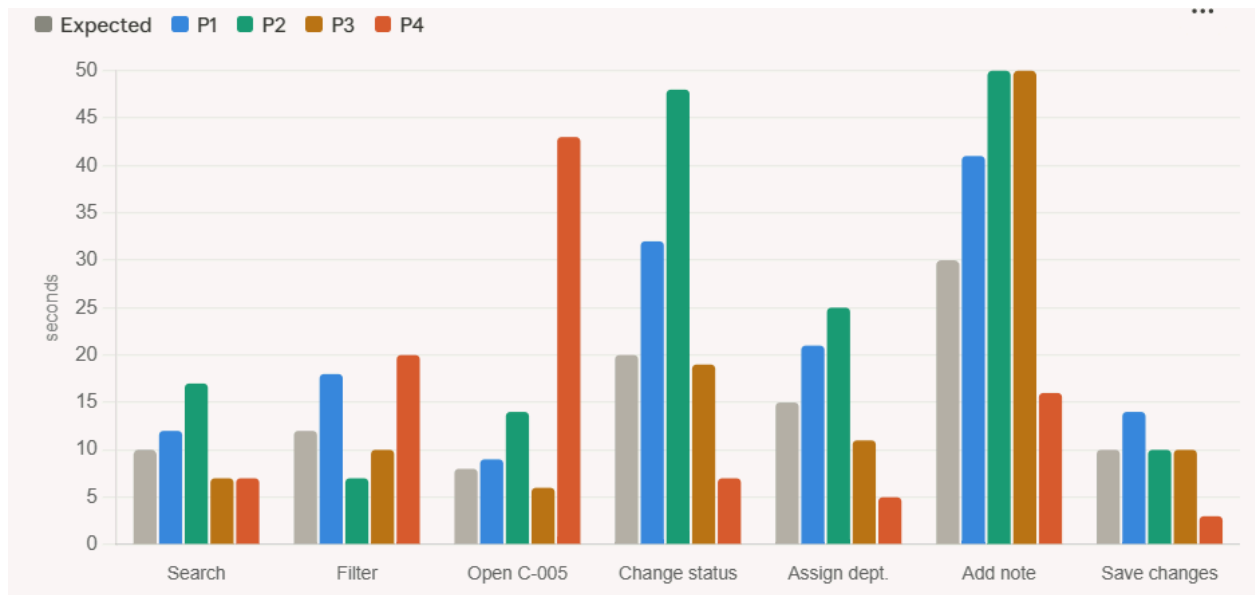


Figure 1: Time per task vs expected — all participants (seconds)

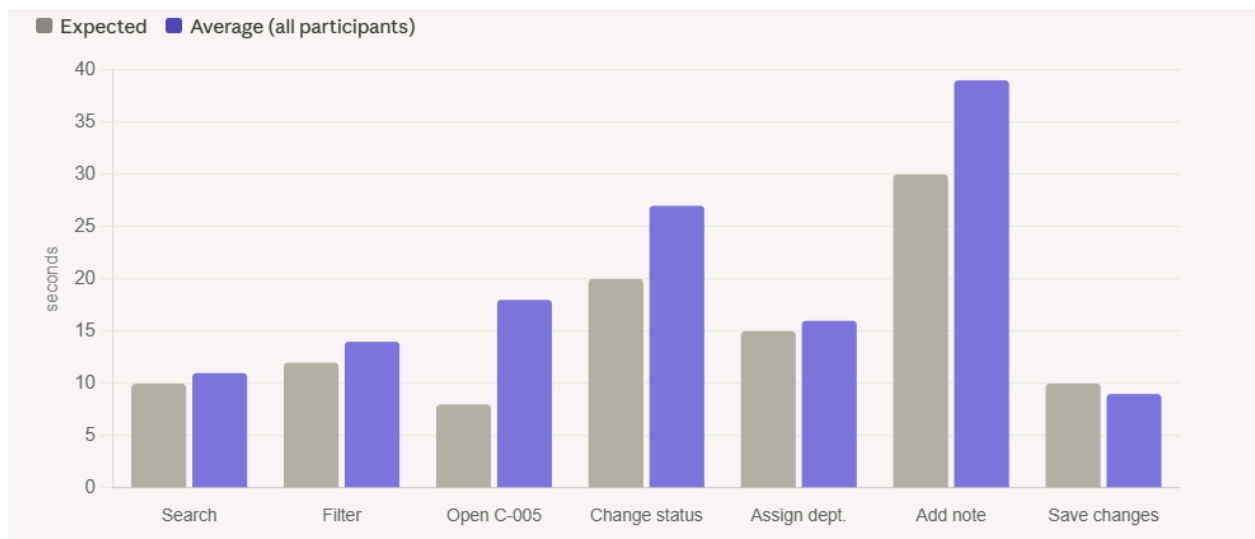


Figure 2: Average time per task vs expected (seconds)

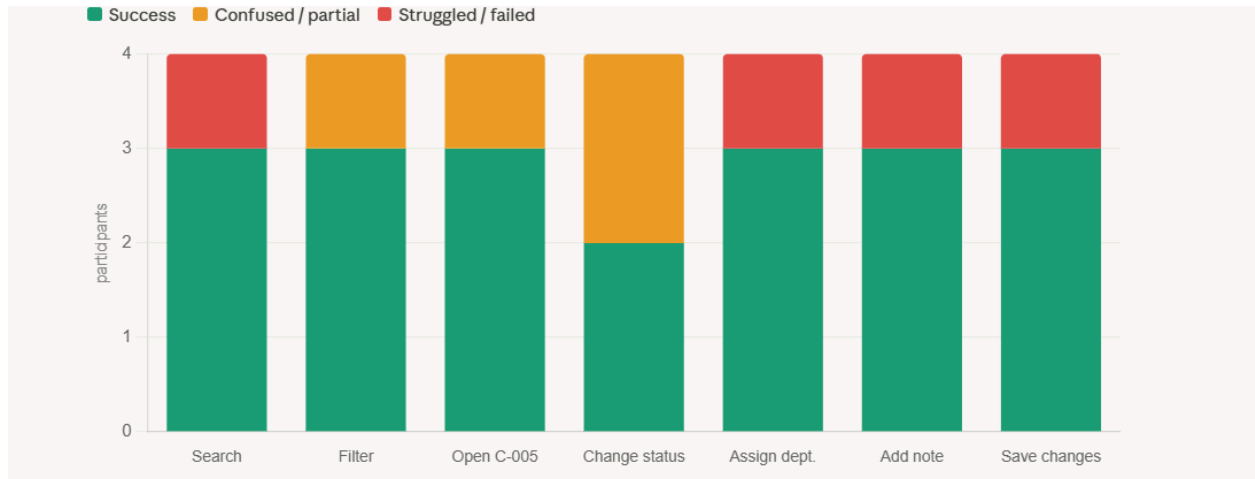


Figure 3: Task outcomes across all 4 participants

Easy ($\leq 1\times$) Moderate ($1-2\times$) Hard ($>2\times$)					
Task	P1	P2	P3	P4	Avg ratio
Search	1.2×	1.7×	0.7×	0.7×	1.1×
Filter	1.5×	0.6×	0.8×	1.7×	1.1×
Open C-005	1.1×	1.8×	0.8×	5.4×	2.3×
Change status	1.6×	2.4×	0.9×	0.3×	1.3×
Assign dept.	1.4×	1.7×	0.7×	0.3×	1.0×
Add note	1.4×	1.7×	1.7×	0.5×	1.3×
Save changes	1.4×	1.0×	1.0×	0.3×	0.9×

Figure 4: Difficulty ratio - time spent relative to expected, per participant per task

Final Note

The interface is largely usable and well received.

Most tasks completed successfully, with the layout described as clean and intuitive by the majority of participants.

Issues worth fixing before next iteration: make a more visible arrow to suggest that the field Update Status is a dropdown, remove Save button and have automatic saving and internal notes need to be easier to find.

A filter reset button was also a repeated ask.

Everything else is minor polish.

Scrum

For Sprint 0, we have successfully covered all the necessary planning and preparation activities required for the project. This includes defining the project scope, objectives, personas, user stories, system requirements, and selecting the appropriate technologies and methodology. Since Sprint 0 is focused on planning rather than execution, the use of a project management tool such as Jira was not necessary at this stage.

- All relevant information has been documented and structured within this report, providing a clear foundation for the upcoming development sprints.

Sprint 1

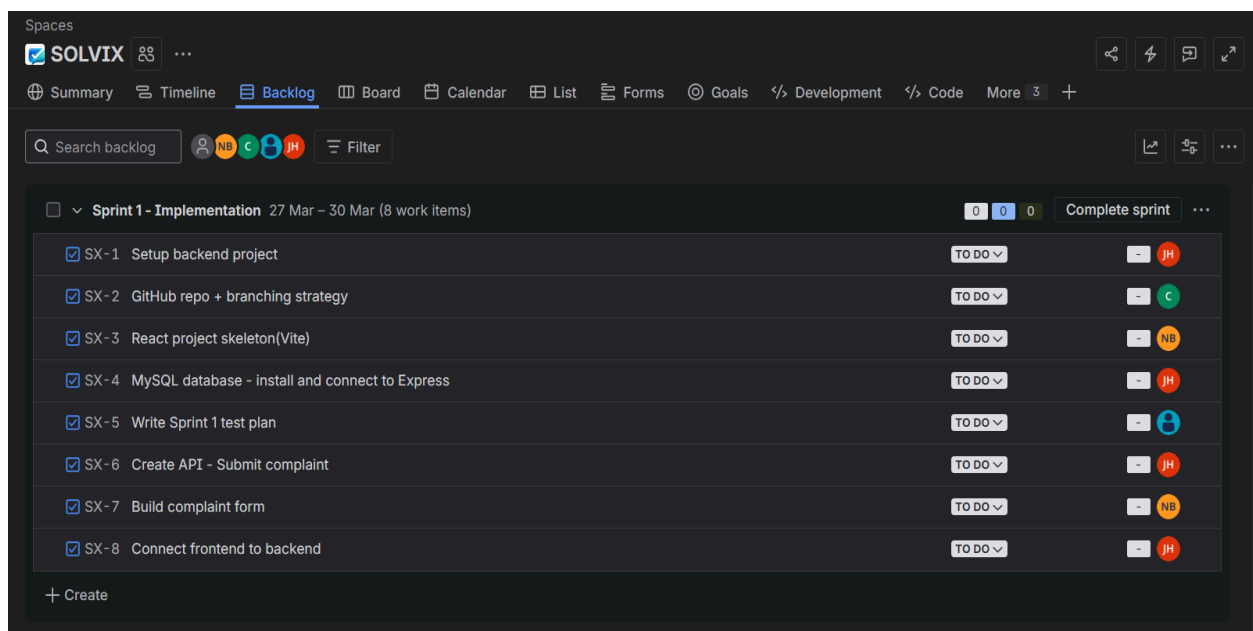


Figure 1 shows the Sprint 1 backlog with all planned tasks

SOLVIX

Summary Timeline Backlog Board Calendar List Forms More 6 +

Search backlog Filter

☐ Sprint 1 - Implementation 27 Mar - 30 Mar (8 work items) 0 0 0 Complete sprint ...

☒ SX-1 Setup backend project	TO DO	JH
☒ SX-2 GitHub repo + branching strategy	TO DO	C
☒ SX-3 React project skeleton(Vite)	TO DO	NB
☒ SX-4 MySQL database - install and connect to Express	TO DO	JH
☒ SX-5 Write Sprint 1 test plan	TO DO	
☒ SX-6 Create API - Submit complaint	TO DO	JH
☒ SX-7 Build complaint form	TO DO	NB
☒ SX-8 Connect frontend to backend	TO DO	JH

+ Create

☐ Backlog (0 work items) 0 0 0 Create sprint

Your backlog is empty.

Jira work item

Add epic / SX-1

Setup backend project

To Do Improve Task

Description

Jaka will initialize Node.js + Express project

Subtasks

Add subtask

Linked work items

Add linked work item

Your pinned fields

Labels

Complexity:medium backend

Figure 2 shows the detailed task view, including assignment, description, and complexity labels.

Burndown chart

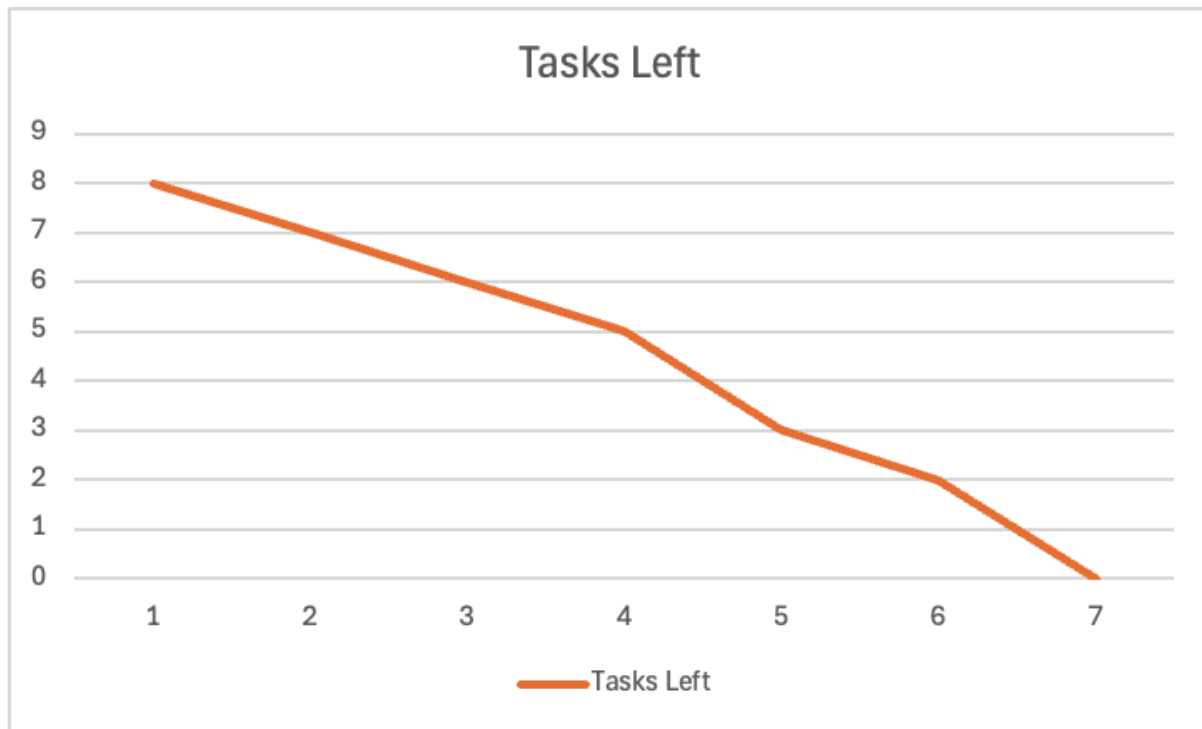


Figure 3: Sprint 1 Burndown Chart: The chart shows the decrease in remaining tasks throughout the sprint, indicating team progress.

Project stakeholders

Symbol	Meaning	Description
+	Ally	Someone who is likely to assist with or advocate for the project.
o	Natural	Someone who isn't against the project but isn't taking an active role in supporting it.
-	Opponent	Someone who is more likely to obstruct or speak out against the project than to back it.

Stakeholder	Status	Expectation	Possible Conflicts or Benefits	Team response
Zalando(Client)	+	Wants an efficient complaint system that improves customer satisfaction and reduces resolution time. Interested in better internal communication and brand reputation.	May demand additional features or changes during development. However, it provides funding (€30,000) and strong project direction.	Maintain regular communication (weekly reports), define clear scope, manage change requests carefully, involve them in key decisions.
Customer Care Manager	+	Wants centralized dashboard, fast complaint handling, better team performance tracking. Needs a user friendly system.	If the system is too complex, adoption may be slow. But provides valuable feedback and ensures system usability.	Focus on UX simplicity, include them in testing, gather feedback during development, and continue based on their needs.
Financial Accountant	+	Wants structured complaint data, especially refunds, to improve financial reporting accuracy and reduce stress during closing periods.	Conflict if system lacks financial visibility or proper data formatting. Benefit: improves financial planning.	Include filters, export features (CSV), and real-time data visibility. Collaborate with them on reporting features.
Customers	+	Want fast, simple complaint submission and ability to track status. Expect quick responses and transparency.	If system is slow or complicated → dissatisfaction. Benefit: higher user satisfaction and trust.	Design simple interface, enable status tracking, allow file uploads, ensure fast response times.
Internal Departments	o	Want clear assignment of complaints and better communication between departments.	Possible resistance to change or adaptation to new system. Benefit: improved workflow efficiency.	Provide training, ensure the system is intuitive, clearly define responsibilities and workflows.

Project Team (Developers, QA, PM)	+	Wants to deliver a successful project, meet deadlines, and achieve good academic results.	Risk of stress, miscommunication, or overload due to deadlines.	Use Agile (Scrum), regular meetings, clear task division, realistic sprint planning.
ZPS – Zveza potrošnikov Slovenije (Sponsor)	+	Wants the system to represent consumer interests, ensure transparency in complaint handling, and improve customer rights protection.	May push for more consumer facing features over internal ones. Benefit: adds credibility and legitimacy to the project.	Keep them informed through final documentation and reports. Ensure the system fairly represents customer needs alongside company needs.

Engineering Management – Leadership

Positive pattern 1: Remove roadblocks

DEFINITION:

“Remove Roadblocks” focuses on helping the team solve obstacles quickly so development can continue efficiently. This pattern encourages proactive communication, collaboration, and fast problem-solving whenever technical, organizational, or communication issues appear.

RELEVANCE TO OUR TEAM:

In our *Solvix* project, we are developing a complaint management platform for Zalando that includes frontend, backend, database integration, dashboards, complaint tracking, and reporting systems. Since multiple team members work on connected features, blockers can easily slow down progress if they are not addressed immediately.

By removing roadblocks quickly, our team can maintain sprint progress, reduce delays, and improve collaboration between frontend, backend, QA testing, and project management tasks. This helps us stay focused on delivering an organized and user-friendly complaint management system.

STRATEGIES TO ADDRESS THIS POSITIVE PATTERN:

Quick Communication:

We discuss blockers during Scrum stand-ups and immediately inform teammates when technical or organizational problems appear.

Team Collaboration:

If one team member struggles with API integration, database setup, or frontend issues, other members help solve the problem instead of letting work stop.

Prioritizing Critical Issues:

Important issues such as backend crashes, database connection problems, or broken complaint tracking features are solved before moving to less important tasks.

Shared Problem Solving:

We use shared discussions and task boards to monitor sprint progress and identify blockers early before they become major delays.

Positive pattern 2: Set Clear Goals

DEFINITION:

“Set Clear Goals” means defining understandable objectives, responsibilities, and expectations for the entire team. Clear goals help team members stay organized, focused, and aligned during project development.

RELEVANCE TO OUR TEAM:

Our Solvix project follows Agile Scrum methodology with sprint planning, milestones, user stories, and testing objectives. Because the project includes multiple features such as complaint submission, status tracking, department assignment, and analytics, every team member must clearly understand their responsibilities and deadlines.

Having clear goals helps our team reduce confusion, improve time management, and ensure that all project requirements are completed on time.

STRATEGIES TO ADDRESS THIS POSITIVE PATTERN:

Sprint Planning:

Before every sprint, we clearly define what tasks need to be completed and assign responsibilities to team members.

User Stories and Test Cases:

We use user stories and detailed test cases to clearly explain expected functionality and project requirements.

Measurable Objectives:

Tasks are divided into smaller achievable goals so progress can be tracked more easily throughout development.

Progress Tracking:

We monitor sprint progress using Scrum boards and burndown charts to ensure deadlines and milestones are respected.

Positive pattern 3: Be a teacher and mentor

DEFINITION:

“Be a Teacher and Mentor” focuses on supporting teammates, sharing knowledge, and helping the team improve together. Strong collaboration and mentoring create a more skilled, confident, and motivated development team.

RELEVANCE TO OUR TEAM:

Since our Solvix team consists of students with different levels of experience in frontend development, backend development, databases, and testing, teamwork and knowledge sharing are extremely important.

Mentoring helps us improve communication, solve problems faster, and ensure that every team member understands both the technical and organizational aspects of the project.

STRATEGIES TO ADDRESS THIS POSITIVE PATTERN:

Knowledge Sharing:

Team members explain technical concepts, tools, and solutions to each other during development.

Collaborative Problem Solving:

When someone encounters difficulties with coding, database integration, or testing, the team works together to find solutions.

Constructive Feedback:

We give feedback respectfully during testing and sprint reviews to help improve both the project and individual skills.

Supporting Growth:

More experienced members help others understand technologies such as React, Node.js, Express, MySQL, and API integration so the entire team can improve together.

Negative pattern 1: Ignore blockers and delays

DEFINITION:

"Ignore Blockers and Delays" describes the failure to address obstacles promptly, allowing small problems to grow into significant setbacks. This anti-pattern emerges when team members avoid reporting issues or assume problems will resolve themselves.

RELEVANCE TO OUR TEAM:

In Solvix, our components are tightly interconnected. An unresolved blocker in the backend or database layer can halt progress across the entire team. With a fixed three-month schedule, even a few days of unaddressed delay can cascade into missed milestones.

STRATEGIES TO AVOID THIS NEGATIVE PATTERN:

Immediate Reporting:

Blockers must be raised as soon as they are identified, not held until the next scheduled meeting.

Escalation Protocol:

If a blocker is unresolved within 24 hours, it is escalated to the Project Manager.

Visible Tracking:

Blockers are flagged on the Jira board so the entire team maintains real-time awareness.

Negative pattern 2: Micromanagement

DEFINITION:

"Micromanagement" occurs when a leader excessively controls team members' work, leaving little room for autonomy or personal ownership, creating unnecessary bottlenecks in the workflow.

RELEVANCE TO OUR TEAM:

Solvix has clearly defined roles across backend, frontend, and QA. If the Project Manager overrides technical decisions rather than trusting each specialist, it slows development and reduces motivation during critical sprint cycles.

STRATEGIES TO AVOID THIS NEGATIVE PATTERN:

Clear Role Boundaries:

Each team member owns their domain and is trusted to complete tasks independently.

Outcome-Focused Leadership:

The Project Manager monitors whether goals are met, not how individual tasks are implemented.

Trust Through Retrospectives:

Sprint reviews provide structured feedback, removing the need for constant interim check-ins.

Negative pattern 3: Poor communication and siloed work

DEFINITION:

"Poor Communication and Siloed Work" refers to a breakdown in information sharing where individuals work in isolation, leading to duplicated effort, integration failures, and misaligned expectations.

RELEVANCE TO OUR TEAM:

Solvix is a full-stack application where frontend, backend, and QA are tightly coupled. If team members make assumptions about each other's implementations without communicating, integration errors and outdated test cases will require costly rework.

STRATEGIES TO AVOID THIS NEGATIVE PATTERN:

Regular Stand-Ups:

Frequent Scrum meetings keep everyone informed of progress, blockers, and dependencies.

Shared Documentation:

API contracts and database schemas are documented and accessible to all team members.

Cross-Role Alignment:

Before starting interconnected features, relevant team members agree on expected inputs and outputs.

Final Application Showcase

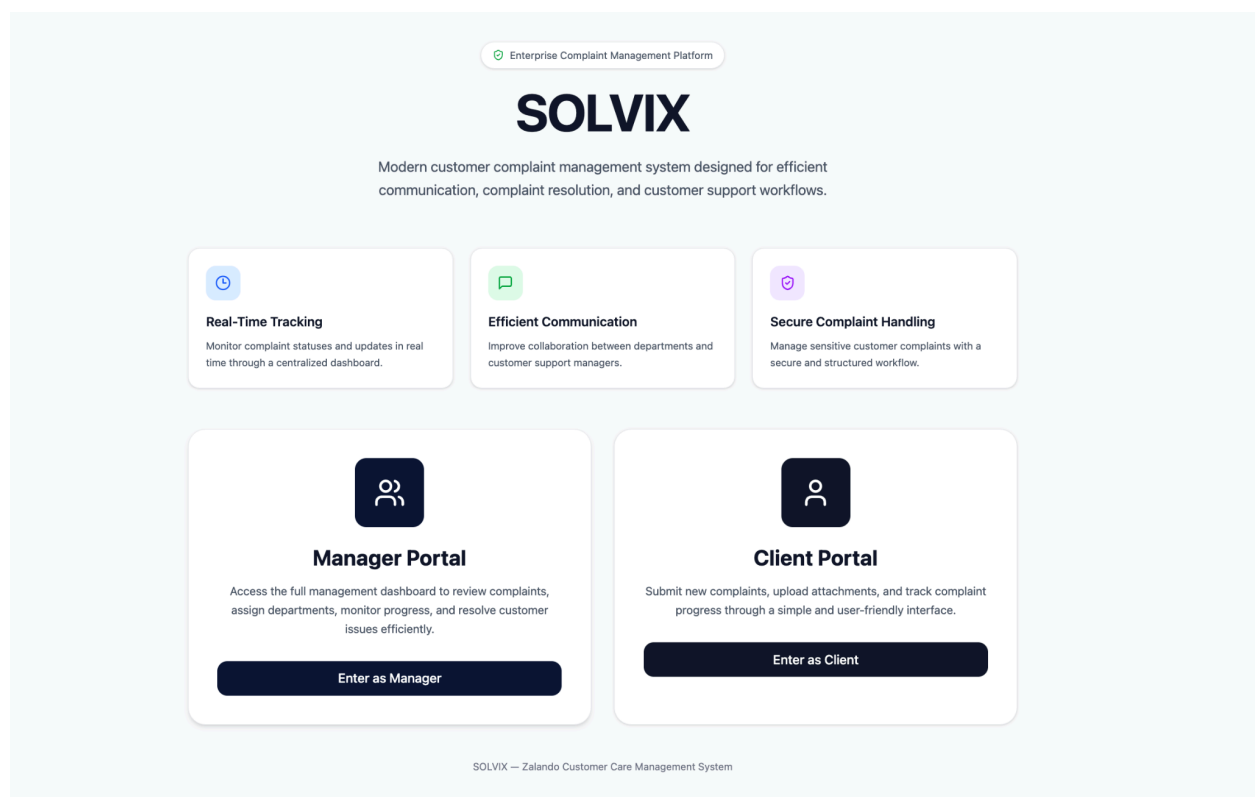
After completing the planning, prototyping, usability testing, and development phases, the following section presents the final Solvix application and its implemented functionality.

The final Solvix application consists of two primary interfaces: the Manager Portal and the Client Portal. The system enables customers to submit and track complaints, while managers can monitor, assign, and resolve complaints through a centralized dashboard.

The application was designed to improve communication between customers and support departments while providing a structured and organized workflow for complaint handling.

Landing Page

The landing page serves as the entry point to the system and provides access to both the Manager Portal and the Client Portal. It also presents the core features of the platform, including real-time tracking, efficient communication, and secure complaint handling.



The page is designed to clearly separate the two user roles:

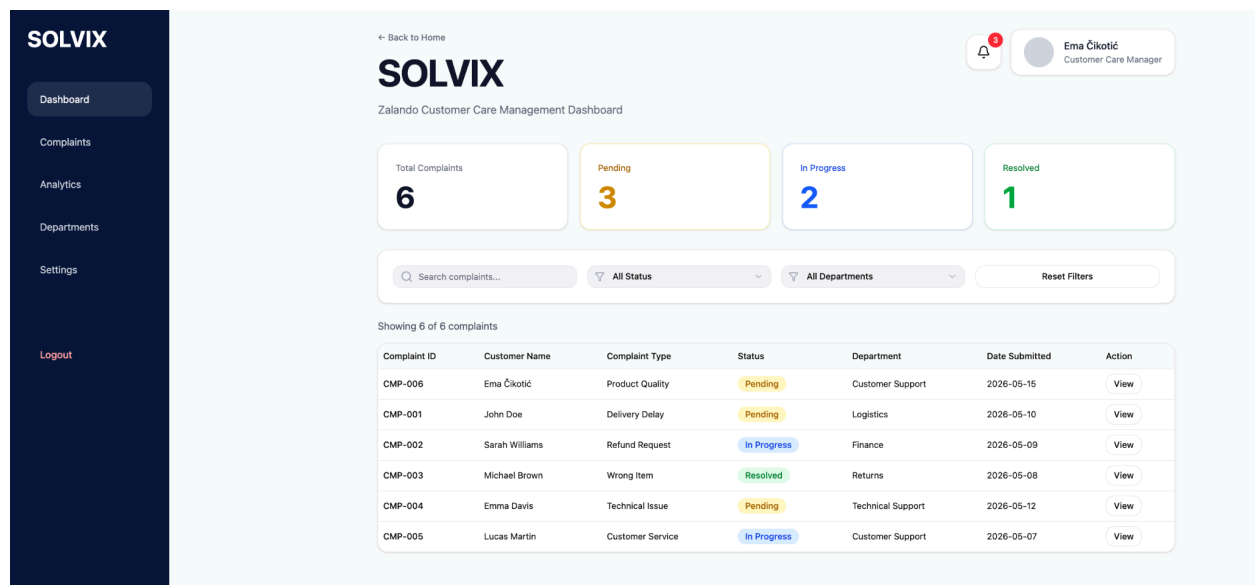
- Manager Portal for employees and customer care managers
- Client Portal for customers submitting complaints

The layout focuses on simplicity and accessibility while introducing the main purpose of the application.

Manager Portal

The Manager Portal allows Customer Care Managers to monitor complaints, assign departments, update statuses, and analyze complaint performance through a centralized dashboard.

Manager Dashboard



The dashboard displays all submitted complaints in a structured table format. Managers can:

- Search complaints
- Filter complaints by status or department
- View complaint details
- Monitor complaint statistics
- Access notifications

The top section of the dashboard contains summary cards showing:

- Total complaints
- Pending complaints
- Complaints in progress

- Resolved complaints

This provides managers with a quick overview of the current complaint situation.

The filtering system helps managers quickly locate specific complaints and improves workflow efficiency when handling larger numbers of complaints.

Complaint Details Page

The screenshot displays the SOLVIX Complaint Management interface. At the top, the header includes the SOLVIX logo, the text 'Complaint Management', a notification bell icon with a red dot, and a user profile for 'Ema Čikotić, Customer Care Manager'. Below the header, a navigation link '← Back to Dashboard' is visible. The main content area is divided into several sections:

- Complaint ID:** CMP-001, with a yellow 'Pending' status tag.
- Customer Information:** Customer Name 'John Doe', Department 'Logistics', and Complaint Type 'Delivery Delay'.
- Submission Date:** 2026-05-10.
- Complaint Description:** A text box containing the message: 'My order was delayed for more than 10 days and I received no updates regarding shipment.'
- Attachments:** A section indicating 'No attachments uploaded'.
- Manage Complaint:** A sidebar containing:
 - Assign Department:** A dropdown menu currently set to 'Logistics'.
 - Update Status:** A dropdown menu currently set to 'Pending'.
 - Save Changes:** A dark blue button.
 - Assigned To:** A section showing 'Assigned To Anna Novak' with a user icon.
 - Last Updated:** A section showing 'Last Updated 2 hours ago' with a clock icon.
- Internal Notes:** A section with a placeholder 'Add notes...' and a text input area.

The Complaint Details page provides detailed information about an individual complaint, including:

- Customer name
- Complaint type
- Assigned department
- Submission date
- Complaint description
- Attachments

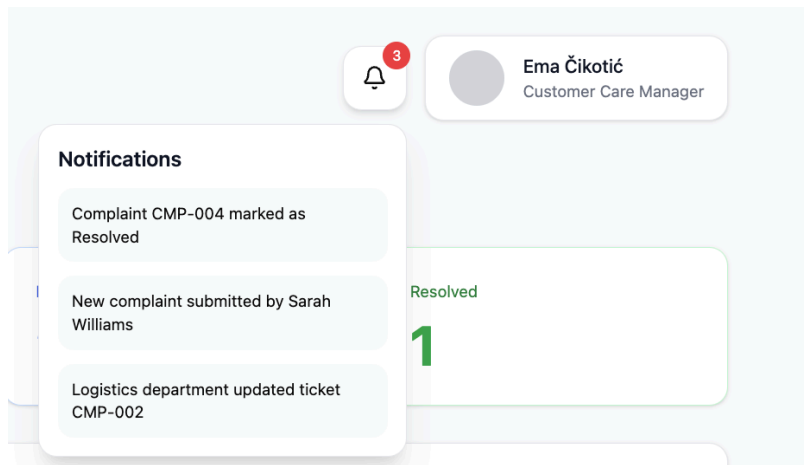
Managers can also:

- Assign complaints to departments
- Update complaint statuses

- Add internal notes
- Save changes

The internal notes section allows employees to communicate internally without exposing information to customers.

Notifications System



The notification system informs managers about important complaint updates in real time.

Examples include:

- Newly submitted complaints
- Status updates
- Complaint resolutions
- Department activity

This helps managers remain informed without constantly refreshing the dashboard.

Analytics Dashboard



The analytics dashboard provides an overview of complaint performance and department activity.

The dashboard includes:

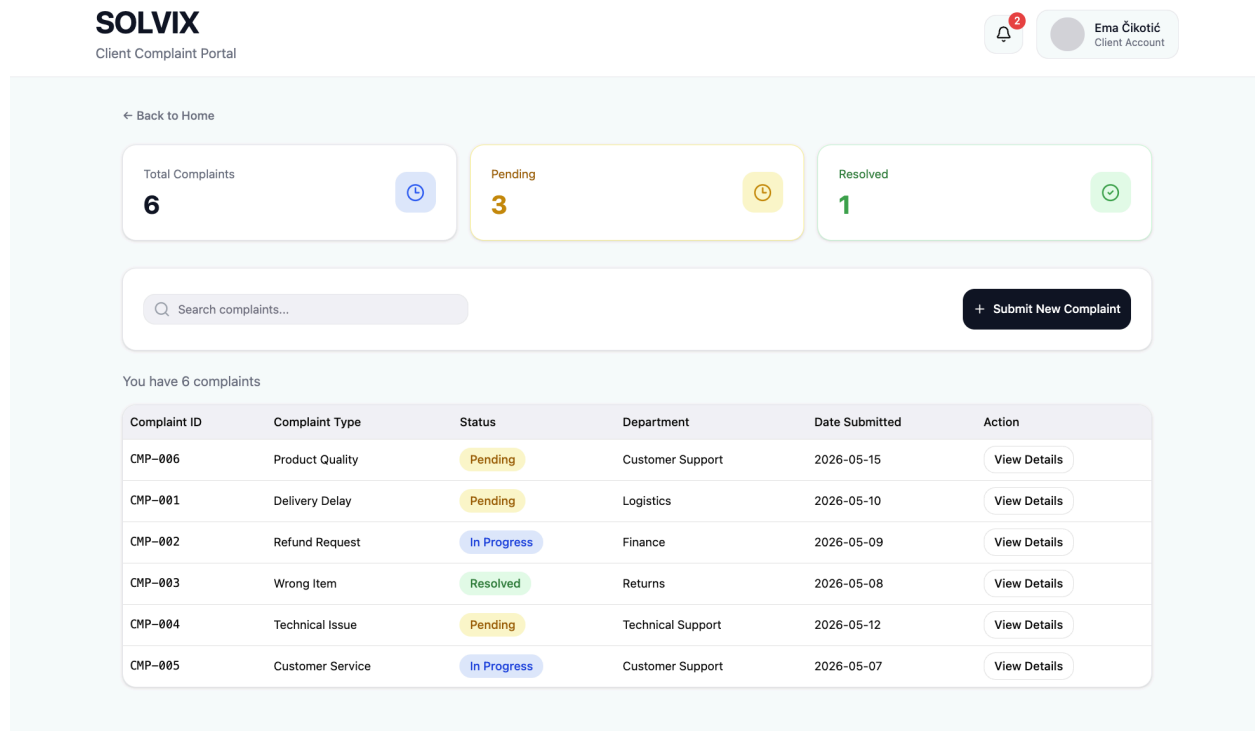
- Resolution rate
- Average response time
- Customer satisfaction metrics
- Complaint distribution by department

The visual statistics help management identify bottlenecks and monitor overall support efficiency.

Client Portal

The Client Portal allows customers to submit complaints, upload attachments, and track complaint progress through a simple and user-friendly interface.

Client Dashboard



The client dashboard displays all complaints submitted by the customer.

Customers can:

- View complaint statuses
- Search complaints
- Open complaint details
- Submit new complaints

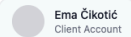
The status system allows customers to immediately see whether a complaint is:

- Pending
- In Progress
- Resolved

This improves transparency and reduces the need for repeated email communication.

Complaint Submission Form

SOLVIX
Submit a New Complaint



[← Back to My Complaints](#)

Submit Complaint

Fill out the form below and our support team will review your complaint shortly.

Full Name

Enter your full name

Complaint Type

Select complaint type

Complaint Description

Describe your issue in detail...

Minimum 20 characters0/500

Attachments


Drag & drop files here
or click to browse files
PDF, PNG, JPG up to 5MB

Submit Complaint

**Quick Review**
Complaints are reviewed within 24–48 hours.

**Live Tracking**
Track complaint status directly from your dashboard.

**File Uploads**
Attach screenshots, invoices, or supporting documents.

Customers can submit complaints through a dedicated complaint form.

The form includes:

- Full name
- Complaint type
- Complaint description
- Attachment upload

The attachment system allows users to upload screenshots or additional evidence related to their issue.

Filled Complaint Example

SOLVIX
Submit a New Complaint

 **Ema Čikotić**
Client Account

[← Back to My Complaints](#)

Submit Complaint

Fill out the form below and our support team will review your complaint shortly.

Full Name

Complaint Type

Ema Čikotić

Product Quality

Complaint Description

The product I received was damaged upon arrival and does not match the description shown on the website. I would like a replacement or refund.

Minimum 20 characters142/500

Attachments



Drag & drop files here
or click to browse files
PDF, PNG, JPG up to 5MB

Submit Complaint

**Quick Review**
Complaints are reviewed within 24–48 hours.

**Live Tracking**
Track complaint status directly from your dashboard.

**File Uploads**
Attach screenshots, invoices, or supporting documents.

Complaint submitted successfully! Your ticket has been created.

[← Back to My Complaints](#)

The form validates user input before submission and ensures complaints contain enough information for support teams to process them efficiently.

After submission, the complaint is automatically stored in the database and becomes visible to managers.

Updated Complaint List

SOLVIX

Client Complaint Portal

2

Ema Ćikotić

Client Account

← Back to Home

Total Complaints

8

Pending

5

Resolved

1

Search complaints...

+ Submit New Complaint

You have 8 complaints

Complaint ID	Complaint Type	Status	Department	Date Submitted	Action
CMP-008	Product Quality	Pending	Customer Support	2026-05-15	View Details
CMP-007	Product Quality	Pending	Customer Support	2026-05-15	View Details
CMP-006	Product Quality	Pending	Customer Support	2026-05-15	View Details
CMP-001	Delivery Delay	Pending	Logistics	2026-05-10	View Details
CMP-002	Refund Request	In Progress	Finance	2026-05-09	View Details
CMP-003	Wrong Item	Resolved	Returns	2026-05-08	View Details
CMP-004	Technical Issue	Pending	Technical Support	2026-05-12	View Details
CMP-005	Customer Service	In Progress	Customer Support	2026-05-07	View Details

Once the complaint is submitted successfully, it immediately appears in the customer dashboard with a Pending status.

This demonstrates the real-time synchronization between the customer and management systems.

Complaint Workflow

The overall complaint workflow functions as follows:

1. A customer submits a complaint through the Client Portal.
2. The complaint is stored in the database.
3. Managers can view the complaint through the Manager Portal.
4. Managers assign the complaint to the appropriate department.
5. Complaint statuses are updated throughout the resolution process.
6. Customers can monitor updates directly from their dashboard.

The workflow ensures transparent communication between customers and internal departments while maintaining organized complaint management.

System Features

The final Solvix application includes the following features:

- Centralized complaint dashboard
- Real-time complaint tracking
- Department assignment system
- Complaint status management
- Internal notes
- Notifications system
- Analytics dashboard
- File attachment uploads
- Search and filtering functionality
- Customer complaint portal

Conclusion

The final Solvix prototype demonstrates a complete complaint management workflow for both customers and customer care managers.

The application focuses on usability, transparency, and efficient complaint handling through centralized dashboards, analytics, notifications, and real-time status tracking.

The final product successfully combines frontend design, backend logic, complaint management functionality, and user-centered workflows into a coherent and practical system.